

# Conic Exercises: Lines, Ellipses, Parabolas, and Tangents

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## §1.1 Line and ellipse

The note continues the conic-section exercises. A line and an ellipse can have three possible positional relationships:

$$\begin{aligned}\Delta > 0 &: \text{two intersections,} \\ \Delta = 0 &: \text{tangent,} \\ \Delta < 0 &: \text{no real intersection.}\end{aligned}$$

This is obtained by substituting the line equation into the conic equation and examining the discriminant of the resulting quadratic.

For example, consider

$$y = 2x - 2, \quad \frac{x^2}{5} + \frac{y^2}{4} = 1.$$

Substitution gives

$$\frac{x^2}{5} + \frac{(2x - 2)^2}{4} = 1,$$

so

$$\frac{6}{5}x^2 - 2x = 0.$$

Thus

$$x = 0, \quad x = \frac{5}{3},$$

and the intersection points are

$$(0, -2), \quad \left(\frac{5}{3}, \frac{4}{3}\right).$$

## §1.2 Tangents from a point to an ellipse

Let

$$C : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

and let  $P = (0, 2)$ . A line through  $P$  has equation

$$y = kx + 2.$$

Substituting into the ellipse gives a quadratic in  $x$ . Tangency occurs exactly when the discriminant is zero.

For the example on the page, this produces

$$k = \pm \frac{\sqrt{6}}{3},$$

so the two tangents are

$$y = \frac{\sqrt{6}}{3}x + 2, \quad y = -\frac{\sqrt{6}}{3}x + 2.$$

### §1.3 Line and hyperbola

For the hyperbola

$$\frac{x^2}{2} - \frac{y^2}{2} = 1,$$

and the line

$$y = x - 1,$$

substitution gives

$$x^2 - (x - 1)^2 = 2.$$

Hence

$$2x - 1 = 2, \quad x = \frac{3}{2}, \quad y = \frac{1}{2}.$$

The line is not tangent; it meets the hyperbola at one point because the equation degenerates to a linear equation after substitution. This is a useful warning: for hyperbolas, “one intersection” need not mean tangency, because the line may be parallel to an asymptotic direction.

### §1.4 Parabola tangent through a point

For the parabola

$$x^2 = \frac{1}{2}y,$$

we have

$$y = 2x^2.$$

A line through  $P = (2, 6)$  may be written

$$y = kx + 6 - 2k.$$

Tangency means that

$$2x^2 = kx + 6 - 2k$$

has a repeated root. Thus

$$2x^2 - kx - 6 + 2k = 0,$$

and the discriminant condition is

$$k^2 - 4 \cdot 2(-6 + 2k) = 0.$$

Solving gives

$$k = 12, \quad k = -4.$$

The tangent lines are therefore

$$y = 12x - 18, \quad y = -4x + 14.$$

## §1.5 A parabola and a circle

The second page considers the parabola

$$C : x^2 = 2py, \quad p > 0,$$

with focus  $F$ , and a circle

$$M : x^2 + (y + 4)^2 = 1.$$

The problem asks for  $p$  when the shortest distance between  $F$  and the circle is 4, and then asks for a maximal triangle area using two tangents from a point  $P$  on the circle.

Since the focus of  $x^2 = 2py$  is

$$F = \left(0, \frac{p}{2}\right),$$

and the circle has center

$$O = (0, -4)$$

and radius 1, the distance from  $F$  to the circle is

$$FO - 1 = \left(\frac{p}{2} + 4\right) - 1 = \frac{p}{2} + 3.$$

Setting this equal to 4 gives

$$p = 2.$$

Hence the parabola is

$$x^2 = 4y.$$

For a tangent point  $A = (x_1, x_1^2/4)$  on  $x^2 = 4y$ , the tangent has equation

$$xx_1 = 2(y + x_1^2/4),$$

or

$$y = \frac{x_1}{2}x - \frac{x_1^2}{4}.$$

Similarly for a second tangent point  $B = (x_2, x_2^2/4)$ . The two tangents meet at

$$P = \left(\frac{x_1 + x_2}{2}, \frac{x_1 x_2}{4}\right).$$

Using the condition that  $P$  lies on the circle gives an optimization problem for the area of triangle  $PAB$ . The computation in the page reduces it to a one-variable expression and obtains the maximum

$$[PAB]_{\max} = 20\sqrt{5}.$$

## §1.6 An ellipse with prescribed quadrilateral area

The final exercise considers an ellipse

$$E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad a > b > 0,$$

passing through  $A = (0, -2)$ . Four vertices form a quadrilateral of area  $4\sqrt{5}$ . Since  $A = (0, -2)$  lies on the ellipse,  $b = 2$ . The area condition gives

$$2ab = 4\sqrt{5},$$

so

$$a = \sqrt{5}.$$

Therefore

$$E : \frac{x^2}{5} + \frac{y^2}{4} = 1.$$

For the line through  $P = (0, -3)$

$$\ell : y = kx - 3,$$

intersecting the ellipse at  $B, C$ , the equation is

$$\frac{x^2}{5} + \frac{(kx - 3)^2}{4} = 1.$$

This gives a quadratic in  $x$ . The condition that the line meet the ellipse in two points is

$$k > 1 \quad \text{or} \quad k < -1.$$

The problem then studies where the chords meet the line  $y = -3$  and imposes

$$|PM| + |PN| \leq 15.$$

The computation on the page reduces this to

$$|k| \leq 3.$$

Combining conditions gives

$$k \in [-3, -1) \cup (1, 3].$$

## §1.7 The next layer

The page is a concentrated exercise in the algebraic method for conics: substitute a line into a conic, read the discriminant, and interpret the result geometrically. The only subtlety is that different conics behave differently: for ellipses, one intersection usually signals tangency; for hyperbolas, a line parallel to an asymptotic direction can produce a single finite intersection without being tangent.

## §1.8 Tangency via discriminants

A line intersecting a conic usually produces a quadratic equation in the line parameter. Tangency occurs when the quadratic has a repeated root:

$$\Delta = 0.$$

This converts a geometric contact condition into algebra. It is the same multiplicity idea that appears in polynomial factorization.

## §1.9 Polarity with respect to a conic

For a nondegenerate conic, a point outside the conic determines a polar line through its contact points with tangents. In matrix form, if the conic is

$$X^T A X = 0,$$

then the polar of  $P$  is

$$P^T A X = 0.$$

This gives a projective explanation of many tangent-chord relations.

### **§1.10 Area constraints and affine invariance**

Some ellipse problems impose an area or quadrilateral condition. Area changes by the determinant of an affine map. Therefore one can often transform an ellipse to a circle, solve the symmetric problem, then transform back with the determinant factor.

### **§1.11 Tangency, polarity, and affine normalization**

Line-conic exercises are controlled by three ideas: intersection multiplicity, polar duality, and affine normalization. The computations are coordinate expressions of these invariant geometric mechanisms.